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RESEARCH INTERESTS

- **Patient-specific computational hemodynamics:** image-based coronary CFD (incl. lattice Boltzmann method) and fractional flow reserve (FFR) assessment
- **Physics-informed machine learning:** neural operators and physics-informed surrogates for real-time cardiovascular flow prediction
- **Image-based cardiovascular physiology:** CT-FFR and diffusion-based 4D coronary CT synthesis for dynamic FFR
- **Wearable hemodynamic diagnostics:** PPG-based estimation of non-Newtonian blood viscosity, blood pressure and fluid status

EDUCATION

Yonsei University

Seoul, Republic of Korea

Ph.D. in Mechanical Engineering (Integrated M.S.–Ph.D. Program)

Mar. 2020 – Aug. 2026

- Dissertation: “Patient-Specific Computational Hemodynamics: Flow Field Analysis for Cardiovascular Assessment”
- Advisor: Prof. Joon Sang Lee

Pusan National University

Busan, Republic of Korea

B.Eng. in Mechanical Engineering

Mar. 2016 – Feb. 2020

RESEARCH EXPERIENCE

Yonsei University

Seoul, Republic of Korea

Postdoctoral Researcher, Multiscale Fluid Dynamics Lab (PI: Prof. Joon Sang Lee)

Sep. 2026 – Present

Appointment: Technical Research Personnel (alternative military service, through Aug. 2027), Engineering Research Institute, Yonsei University

Concurrent: Project Researcher, Yonsei University Health System – 4D CT-FFR project (PI: Prof. Jung-Sun Kim, Department of Internal Medicine, College of Medicine)

- Physics-informed neural operators for real-time, patient-specific prediction of coronary flow fields (journal manuscript in preparation).
- Wearable PPG biomarkers: deep-learning ensemble that predicts fluid-bolus needs from smartwatch PPG spectrograms [M1]; blood pressure, viscosity and hydration estimation.
- Dynamic 4D CT-FFR and 4D flow-based coronary risk assessment in government-funded projects (KHIDI, NRF).

Graduate Research Assistant, Multiscale Fluid Dynamics Lab (Advisor: Prof. Joon Sang Lee) *Mar. 2020 – Aug. 2026*

- Coronary CT-FFR: built image-based, patient-specific CFD pipelines (incl. lattice Boltzmann method) and a gray-zone FFR prediction algorithm combining hemodynamic features, synthetic vessel models and biometric data [J4].
- Physics-informed neural operators: developed a PINO surrogate (PointNet++ branch, Fourier-feature trunk, Navier–Stokes residual regularization) that predicts 3D coronary pressure and velocity fields on unseen geometries in seconds [C1].
- Dynamic 4D CT-FFR: contributed to a diffusion deformable model that synthesizes full-cycle 4D coronary CT from two CT phases for time-resolved FFR analysis [J1, C2].
- Wearable hemodynamics: designed a physics-constrained CNN-LSTM with a Carreau–Yasuda prior to estimate patient-specific non-Newtonian blood viscosity from smartwatch PPG [J2].
- Cardiac device hemodynamics: CFD-based optimization of a tricuspid regurgitation membrane device [J3].

E8 (E8IGHT Co., Ltd.)

Seoul, Republic of Korea

Research Intern

Jan. 2021 – Jun. 2021

- Performed lattice Boltzmann method (LBM)-based computational fluid dynamics simulations.

PUBLICATIONS

† first author; * corresponding author

Journal Articles (SCIE)

- [J1] T.H. Han†, Y.W. Kim, **H.J. Lee**, J.S. Kim, S.G. Lee, D.H. Yang, H.M. Oh, D.S. Kim, S.Y. Shin, S. Song, J.S. Lee*, “Synthesis of coronary 4D CT image by denoising diffusion probabilistic model,” *Computer Methods and Programs in Biomedicine*, vol. 282, 109382, 2026. doi:10.1016/j.cmpb.2026.109382

- [J2] **H.J. Lee**†, Y.W. Kim, S.Y. Shin, S.L. Lee, C.H. Kim, K.S. Chung*, J.S. Lee*, “A Physics-Integrated Deep Learning Approach for Patient-Specific Non-Newtonian Blood Viscosity Assessment using PPG,” *Computer Methods and Programs in Biomedicine*, vol. 265, 108740, 2025. doi:10.1016/j.cmpb.2025.108740
- [J3] Y.W. Kim†, **H.J. Lee**, S.J. Jung, J.H. Kim*, J.S. Lee*, “Optimization of tricuspid membrane mechanism for effectiveness and leaflet longevity through hemodynamic analysis,” *Engineering Applications of Computational Fluid Mechanics*, vol. 16, no. 1, pp. 1587–1600, 2022. doi:10.1080/19942060.2022.2104929
- [J4] **H.J. Lee**†, Y.W. Kim†, J.H. Kim, Y.J. Lee, J.S. Moon, P. Jeong, J.H. Jeong, J.S. Kim*, J.S. Lee*, “Optimization of FFR prediction algorithm for gray zone by hemodynamic features with synthetic model and biometric data,” *Computer Methods and Programs in Biomedicine*, vol. 220, 106827, 2022. doi:10.1016/j.cmpb.2022.106827

Manuscript Under Revision

- [M1] **H.J. Lee**†, J. Myung, J.Y. Hong*, J.S. Lee*, “Multi-Model Deep-Learning Ensemble for Predicting 24-Hour Fluid Bolus Administration from Continuous Photoplethysmography Spectrogram Analysis,” *Engineering Applications of Artificial Intelligence*, first revision under review.

Peer-Reviewed Conference Proceedings

- [C1] **H.J. Lee**†, T.H. Han, J.S. Kim, S.G. Lee, J.S. Lee*, “Patient-Specific Coronary Flow Field Prediction Using Physics-Informed Neural Operators,” *Proc. ASME 2026 Design of Medical Devices Conference (DMD2026)*, Minneapolis, MN, USA, V001T02A010, 2026. doi:10.1115/DMD2026-1091
- [C2] T.H. Han†, Y.W. Kim, **H.J. Lee**, J.S. Kim, S.G. Lee, D.H. Yang, H.M. Oh, J.S. Lee*, “Deriving Dynamic Fractional Flow Reserve From 4D Coronary CT Synthesized by a Diffusion Model,” *Proc. ASME 2026 Design of Medical Devices Conference (DMD2026)*, Minneapolis, MN, USA, V001T02A011, 2026. doi:10.1115/DMD2026-1092

CONFERENCE PRESENTATIONS

International

- [I1] **H.J. Lee** et al., “Patient-Specific Coronary Flow Field Prediction Using Physics-Informed Neural Operators,” ASME Design of Medical Devices Conference (DMD2026), Minneapolis, MN, USA, 2026 (poster).
- [I2] T.H. Han et al. (incl. **H.J. Lee**), “Dynamic Fractional Flow Reserve from 4D Coronary CT Synthesized by a Diffusion Model,” ASME Design of Medical Devices Conference (DMD2026), Minneapolis, MN, USA, 2026 (poster).
- [I3] **H.J. Lee** et al., “Patient-Specific Coronary Flow Field Prediction Using Physics-Informed Neural Operators,” 18th Asian Congress of Fluid Mechanics (ACFM), Seoul, Korea, 2025 (oral).
- [I4] **H.J. Lee** et al., “AI-based Hemorheology Prediction with Patient-Specific Biometric Boundary Conditions,” 26th International Congress of Theoretical and Applied Mechanics (ICTAM), Daegu, Korea, 2024 (short oral & poster).
- [I5] **H.J. Lee**, J.S. Lee, “Optimization of Artificial Intelligence Algorithms for FFR Prediction in Gray Zone,” 10th International Conference of Manufacturing Technology Engineers (ICMTE), Gyeonggi-do, Korea, 2022 (oral).
- [I6] **H.J. Lee** et al., “Estimating CFD-based CT FFR using lattice Boltzmann method – 3D geometry auto segmentation and novel patient-specific computation,” ESCHM-ISCH-ISB 2021, Fukuoka, Japan, 2021 (oral).

Domestic (Korea)

- [D1] **H.J. Lee** et al., “A Physics-Integrated Deep Learning Approach for Patient-Specific Non-Newtonian Blood Viscosity Assessment using PPG,” Biomedical Engineering Society for Circulation (BESCO) Summer Meeting, Gwangmyeong, 2025 (poster).
- [D2] T.H. Han et al. (incl. **H.J. Lee**), “Time-resolved Coronary Flow Visualization from Single-phase CT,” BESCO Summer Meeting, Gwangmyeong, 2025 (poster).
- [D3] D.Y. Lee et al. (incl. **H.J. Lee**), “Arterial Blood Pressure and Blood Viscosity Estimation from AI-based PPG Analysis,” BESCO Winter Meeting, Seongnam, 2024 (poster).
- [D4] S.W. Kwon et al. (incl. **H.J. Lee**), “FFR Prediction with Quantum Machine Learning: Accurate and Non-invasive Diagnosis with Limited Features,” BESCO Winter Meeting, Seongnam, 2024 (poster).
- [D5] D.Y. Lee et al. (incl. **H.J. Lee**), “Blood viscosity estimation from wearable device using AI-based PPG analysis,” BESCO Summer Meeting, Daejeon, 2024 (poster).
- [D6] S.E. Kim et al. (incl. **H.J. Lee**), “Carved membrane design for tricuspid valve regurgitation treatment,” BESCO Summer Meeting, Daejeon, 2024 (poster).
- [D7] **H.J. Lee**, J.S. Lee, “Unlocking Predictive Health Outcomes with Biometric Data,” Korean Society of Mechanical Engineers (KSME) Conference, Songdo, 2023 (oral).

- [D8] **H.J. Lee** et al., “Modeling Coronary Artery Hemodynamics: Exploring DCNN Surrogate Models in Preliminary Research,” BESCO Summer Meeting, Daegu, 2023 (oral).
- [D9] **H.J. Lee** et al., “Artificial Intelligence Algorithms for FFR Prediction in Gray Zone by Single-view Angiography,” BESCO Winter Meeting, Seoul, 2022 (short oral & poster; Best Paper Award nominee).
- [D10] J.H. Kim et al. (incl. **H.J. Lee**), “Autonomous Cardiovascular Hemodynamics Analysis using Patient-Specific CFD Simulation with Unsupervised Learning-based Coronary Artery Segmentation,” BESCO Summer Meeting, Seoul, 2021 (oral).

PATENTS

16 applications in total (12 Korea, 2 US, 2 PCT); 2 registered.

Registered

- [P1] J.S. Lee, M.Y. Yoon, S.C. Ko, **H.J. Lee**, “AI-based noninvasive urodynamics test method and apparatus,” Korean Patent No. [10-2902500](#), registered Dec. 2025 (Appl. No. 10-2022-0085583).
- [P2] J.S. Lee, **H.J. Lee**, Y.W. Kim, “Optimization system and method of AI algorithm for prediction coronary artery lesions based on FFR,” Korean Patent No. [10-2882193](#), registered Nov. 2025 (Appl. No. 10-2022-0030019).

Pending

- [P3] J.S. Lee, J.S. Kim, **H.J. Lee**, S.G. Lee, “Patient-specific coronary flow prediction method based on physics-informed neural operators,” KR Appl. No. 10-2026-0003225.
- [P4] J.S. Lee, J.S. Kim, S.G. Lee, J.H. Kim, Y.W. Kim, **H.J. Lee**, “Deep learning-based multiphase coronary CT interpolation system and method,” US Appl. No. 19/397,163.
- [P5] J.S. Lee, J.S. Kim, S.G. Lee, J.H. Kim, Y.W. Kim, **H.J. Lee**, “Deep learning-based multiphase coronary CT interpolation system and method,” KR Appl. No. 10-2025-0149607.
- [P6] J.S. Lee, **H.J. Lee**, “Device and method for predicting viscosity using AI in a viscosity prediction system,” PCT/KR2025/002462.
- [P7] J.S. Lee, **H.J. Lee**, “Device and method for predicting viscosity using AI in a viscosity prediction system,” KR Appl. No. 10-2025-0021284.
- [P8] J.S. Lee, J.S. Kim, S.G. Lee, J.H. Kim, Y.W. Kim, **H.J. Lee**, “Deep learning-based multiphase coronary CT interpolation method,” KR Appl. No. 10-2024-0168733.
- [P9] J.S. Lee, **H.J. Lee**, “Method for predicting blood glucose and diabetes using PPG signals and biometric information,” KR Appl. No. 10-2024-0147541.
- [P10] J.S. Lee, W.R. Choi, **H.J. Lee**, “Wearable device for monitoring glaucoma suspect and method for monitoring glaucoma suspect,” PCT/KR2024/012365.
- [P11] J.S. Lee, **H.J. Lee**, “Non-Newtonian fluid viscosity modeling of patient blood using wearable device-based PPG and biometric information,” KR Appl. No. 10-2024-0026709.
- [P12] J.S. Lee, **H.J. Lee**, “Systolic and diastolic viscosity prediction algorithm using wearable device-based PPG and biometric information,” KR Appl. No. 10-2024-0028201.
- [P13] J.S. Lee, **H.J. Lee**, “Glucose and diabetes prediction algorithm using wearable device-based PPG and biometric information,” KR Appl. No. 10-2023-0144347.
- [P14] J.S. Lee, W.R. Choi, **H.J. Lee**, “Wearable device for monitoring glaucoma suspect and method for monitoring glaucoma suspect,” KR Appl. No. 10-2023-0118582.
- [P15] J.S. Lee, **H.J. Lee**, “Glaucoma diagnosis method and system based on contactless biosignals,” KR Appl. No. 10-2023-0078883.
- [P16] J.S. Lee, **H.J. Lee**, “Optimization system and method of AI algorithm for prediction coronary artery lesions based on FFR,” US Appl. No. 17/820,819.

RESEARCH PROJECTS (PARTICIPATING RESEARCHER)

- [R1] “Real-Time Precision Medicine Platform Based on Smart Hemodynamic Indices,” National Research Foundation of Korea (NRF), Leading Research Center Program (NRF-2022R1A5A1022977), Jun. 2022 – Feb. 2029.
- [R2] “4D Coronary Flow Visualization from CT in Real Time (4D CT-FFR),” Ministry of Health and Welfare / KHIDI, Clinical-Needs Translational Research (RS-2024-00406717), Apr. 2024 – Dec. 2026.
- [R3] “4D Flow-Based Diagnostic Device for Dynamic Risk Assessment of Anomalous Origin of the Coronary Artery in Children and Adolescents,” Ministry of Health and Welfare / KHIDI, Pediatric Disease R&D Program (RS-2025-02263580), Apr. 2025 – Dec. 2028.
- [R4] “Patient-Specific Coronary Artery Disease Risk Analysis Software,” Ministry of SMEs and Startups, Startup Growth Technology Development Program (TIPS) (RS-2024-00441761), Jul. 2024 – Jun. 2026.

- [R5] “Patient-Specific Implantable Device for Tricuspid Regurgitation Using 3D Heart Printing and 4D-CFD Simulation,” Ministry of Health and Welfare / Korea Medical Device Development Fund (KMDF_PR_20200901_0104), Sep. 2020 – Dec. 2025.
- [R6] “Variable Negative-Pressure Tunnel for Respiratory Infection Control,” Ministry of Health and Welfare / KHIDI, Infectious Disease Medical Safety Technology Program (HG22C0001), Apr. 2022 – Dec. 2024.
- [R7] “High-Speed Point-of-Care FFR Simulator for Cardiovascular Diagnosis and Treatment,” NRF Bio & Medical Technology Development Program (2017M3A9E907337122), Feb. 2020 – May 2022.

AWARDS AND HONORS

Merit Academic Paper Award , School of Mechanical Engineering, Yonsei University (for [J2])	2026
Best Paper Award , Biomedical Engineering Society for Circulation (BESCO)	2022
Merit Academic Paper Award , Yonsei University	2022
Best Paper Award , School of Mechanical Engineering, Yonsei University	2022
<i>Fellowships and Scholarships</i>	
BK21 FOUR Graduate Research Scholarship (7 semesters), Ministry of Education / NRF	2022 – 2025
BK21 PLUS Scholarship , Ministry of Education / NRF	2020 – 2021
Academic Research Fellowship (ARF) , Yonsei University	2021

TEACHING EXPERIENCE

Teaching Assistant , School of Mechanical Engineering, Yonsei University	Mar. 2020 – Dec. 2020
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SKILLS

Computational modeling: image-based, patient-specific coronary CFD; lattice Boltzmann method; hemodynamic indices (FFR, wall shear stress); non-Newtonian blood rheology

Machine learning: physics-informed neural networks and neural operators (DeepONet/PINO); surrogate modeling; diffusion models; CNN/LSTM; ensemble learning

Biomedical data: coronary CT angiography (incl. 4D CT), invasive FFR, smartwatch PPG signals; multi-institutional clinical collaborations

Languages: Korean (native), English (professional working proficiency)

REFERENCES

Prof. Joon Sang Lee, Professor, School of Mechanical Engineering, Yonsei University (Ph.D. advisor and current PI)
joonlee@yonsei.ac.kr

Additional references available upon request.